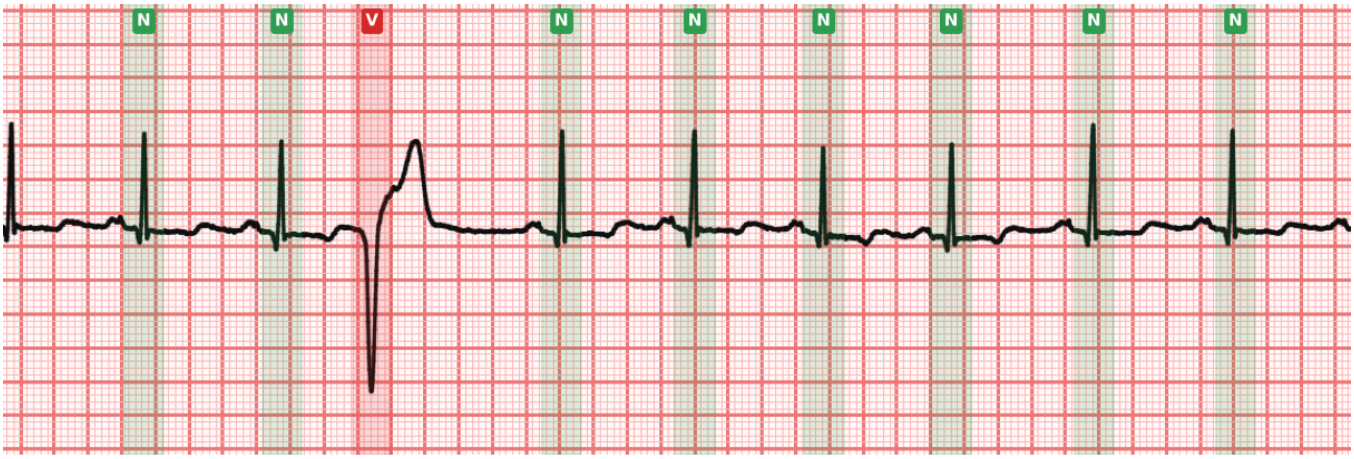
# ECG Arrhythmia Analysis Report

Generated 31 Oct 2026, 00:11 | Source image: 02_isolated_pvcs.png | Analysis mode: calibrated

**9 beats | heart rate 74 bpm | N 8 S 0 V 1**




## Findings

9 beats over 6.48 s (N 8, S 0, V 1, F 0). Heart rate 74.1 bpm on average (5-beat range 70.1-75.5 bpm). 1 PVC(s): 1 isolated, 0 couplet(s). Pattern description of detected beats; not a diagnosis.

## Rhythm statistics

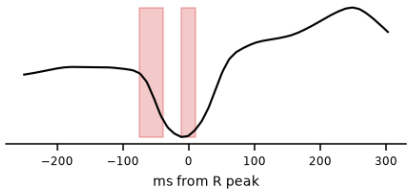
| Measure | Value |
|---|---|
| Beats analysed | 9 |
| Duration | 6.48 s |
| Heart rate (mean) | 74.1 bpm |
| Heart rate (5-beat range) | 70.1 - 75.5 bpm |
| Normal / Supraventricular / Ventricular beats | 8 / 0 / 1 |
| PVC patterns | 1 isolated, 0 couplet(s), 0 run(s) of 3+, 0 bigeminy, 0 trigeminy |
| Beats below confidence threshold | 0 (threshold 55%) |

## Beat-by-beat

| # | Class | Confidence | R-R before | Evidence (short) |
|---|---|---|---|---|
| 1 | Normal | 77% | 0.82 s | Normal-type morphology (confidence 77%); QRS width in line with this strip's other normal beats, timing regular. |
| 2 | Normal | 69% | 0.82 s | Normal-type morphology (confidence 69%); QRS width in line with this strip's other normal beats, timing regular. |
| 3 | Ventricular (PVC) | 77% | 0.53 s | QRS is about 1.7x as wide as this strip's normal beats, consistent with a ventricular origin. |
| 4 | Normal | 99% | 1.12 s | Normal-type morphology (confidence 99%); QRS width in line with this strip's other normal beats, timing regular. |
| 5 | Normal | 90% | 0.79 s | Normal-type morphology (confidence 90%); QRS width in line with this strip's other normal beats, timing regular. |
| 6 | Normal | 74% | 0.76 s | Normal-type morphology (confidence 74%); QRS width in line with this strip's other normal beats, timing regular. |
| 7 | Normal | 72% | 0.76 s | Normal-type morphology (confidence 72%); QRS width in line with this strip's other normal beats, timing regular. |
| 8 | Normal | 86% | 0.84 s | Normal-type morphology (confidence 86%); QRS width in line with this strip's other normal beats, timing regular. |

| # | Class | Confidence | R-R before | Evidence (short) |
|---|---|---|---|---|
| 9 | Normal | 82% | 0.84 s | Normal-type morphology (confidence 82%); QRS width in line with this strip's other normal beats, timing regular. |

## Evidence for flagged beats (Grad-CAM = red shading on the beat; rule-based reasons)




**Beat 3: Ventricular (PVC)**
• CNN: ventricular-type morphology (confidence 77%).
• QRS is about 1.7x as wide as this strip's normal beats, consistent with a ventricular origin.
• No discernible P wave detected before the QRS complex (suppressed/absent P wave is a hallmark of a PVC's ectopic ventricular origin).
• T wave is deflected opposite to the main QRS direction (discordant T wave), typical of abnormal ventricular repolarization in PVC.
• Premature: arrived 21% earlier than the recent R-R average.
• Followed by a pause (66% longer than usual), typical of a compensatory pause.

## Limitations

This report describes patterns found in an ECG image by an automated system trained on the public MIT-BIH Arrhythmia Database (evaluated patient-wise on unseen records). Beats read from an image are less reliable than from a recorded signal; supraventricular (PAC-type) detection is the weakest class, and class F (fusion) is not reported. Beats under the confidence threshold are marked Uncertain instead of being forced into a class. **This is a student research prototype, not a medical device, and not a diagnosis.**